

1. INTRODUCTION

About The Project:

Human Resource management has evolved significantly from a purely administrative function into a strategic component of organizational success. In modern organizations, employees are not just a workforce but a critical asset that directly influences productivity, innovation, and long term growth. Workforce behavior reflect show employees interact with their work environment, how they perform under different conditions, and how they respond to organizational policies. Understanding these patterns is essential for building a stable and efficient organization. Organizations today operate in dynamic environments where employee expectations, market demands, and operational challenges constantly change. As a result, analyzing workforce trends has become an important activity for planning and decision making.

The background of this domain lies in the increasing need for organizations to manage large volumes of employee related data effectively. Earlier, workforce decisions were often based on intuition, limited reports, or manual records. However, with the growth of digital systems, organizations now generate detailed data related to employee demographics, job roles, compensation structures, performance ratings, work experience, and satisfaction levels. This data, when properly analyzed, provides valuable insights into workforce behaviour. It helps organizations understand patterns such as which departments have higher employee retention, which roles experience frequent turnover, and how employee satisfaction varies across different conditions. These insights are essential for maintaining workforce stability and improving overall efficiency.

The importance of this problem becomes clear when considering the impact of employee attrition and workforce imbalance. High employee turnover leads to increased recruitment costs, training expenses, and disruption in workflow. It also affects team morale and reduces organizational productivity. On the other hand, poor workforce planning can result in overstaffing or understaffing, both of which negatively affect operational performance.

Understanding workforce trends allows organizations to identify the root causes behind these issues. For example, factors such as low job satisfaction, inadequate compensation, lack of career growth, or mismatched job roles can contribute to employee attrition. By identifying these factors through data analysis, organizations can take proactive measures to improve employee retention and satisfaction.

Another important aspect is the diversity of employee characteristics within an organization. Employees differ in terms of age, experience, skills, and job roles. These differences influence how individuals perform, how they respond to workplace conditions, and how long they remain with the organization. Analyzing these characteristics helps in understanding how different groups behave and what strategies are required to manage them effectively. For instance, younger employees may prioritize learning opportunities, while experienced employees may focus on stability and compensation. Recognizing such patterns helps organizations design better policies and improve workforce engagement.

Data analytics plays a crucial role in addressing these challenges. By using tools such as pandas and numpy resulting in accurate datasets for further analysis. Once the data is prepared, it is stored in structured formats such as SQL databases. SQL enables efficient querying and analysis of workforce data, allowing users to extract meaningful information based on specific conditions. For example, queries can be used to analyze attrition rates across departments, compare salary distributions, or identify trends in employee performance

2. LITERATURE SURVEY

Python:

Python is a versatile, high-level programming language known for its simplicity, readability, and versatility. Created by Guido van Rossum and first released in 1991, Python has gained widespread popularity and has become one of the most widely used programming languages in various domains, including web development, data analysis, artificial intelligence, scientific computing, and more.

Python is also being used in mission critical applications in the world's largest stock exchange, forms the basis for high end newspaper websites, runs on millions of cell phones, and is used in industries as diverse as ship building, feature length movie animation, and air traffic control. It is a rapidly growing open-source programming language. It is available for most operating systems, including Windows, UNIX, Linux, and Mac OS.

Just like any other programming language, Python vocabulary is actually very small. This is in contrast to the huge vocabulary in any natural language. This small “vocabulary” is called the set of “reserved words”(keywords) in python. These words contain very special meaning to the Python. When Python sees these words in a program, they have a unique meaning to Python. Later as you write programs you will make up your own word that have meaning to your program called variables. You have many options in choosing names for your variables.

Python is a popular high-level programming language used for general-purpose programming, created by Guido van Rossum and first released in 1991. It has wide range of applications such as web development, scientific and mathematical computing, network programming, desktop graphical user Interfaces etc.

The Python Programming Language

Python is a popular programming language. It is used for:

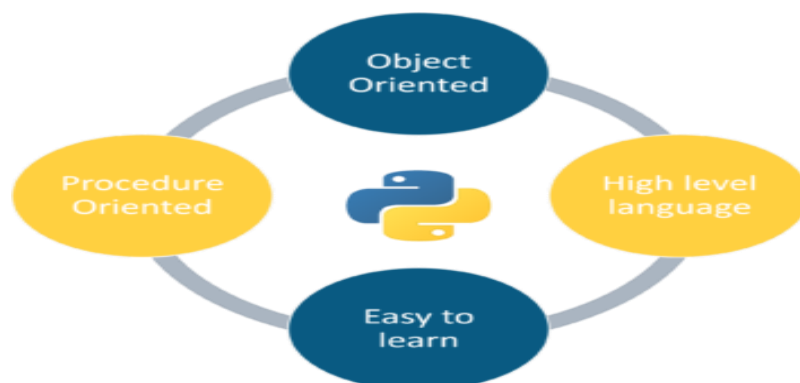
- Web development (server-side)
- Software development.
- Mathematics.

- System scripting.
- Python can be used on a server to create web applications.
- Python can be used a long side software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid proto typing, or for production-ready software development.

Features of Python Programming:

Python's features include:

- Python is intended to be a simple and highly readable language. It is designed to have an uncluttered visual layout, often using English keywords where other languages use punctuation.
- It is a free and open-source software where you can freely use and distribute even for commercial use.
- It ensures portability by providing you to move a Python program from one platform to another, and run it without any changes.
- It is a high-level and interpreted language where you don't have to worry about memory management, garbage collection and so on.
- A large number of standard libraries are available to solve common tasks, which makes life of a programmer much easier since you don't have to write all the code yourself.



The Python source code goes through the following to generate an executable code:

Step 1: The Python compiler reads a Python source code or instruction in the code editor. In this first stage, the execution of the code starts.

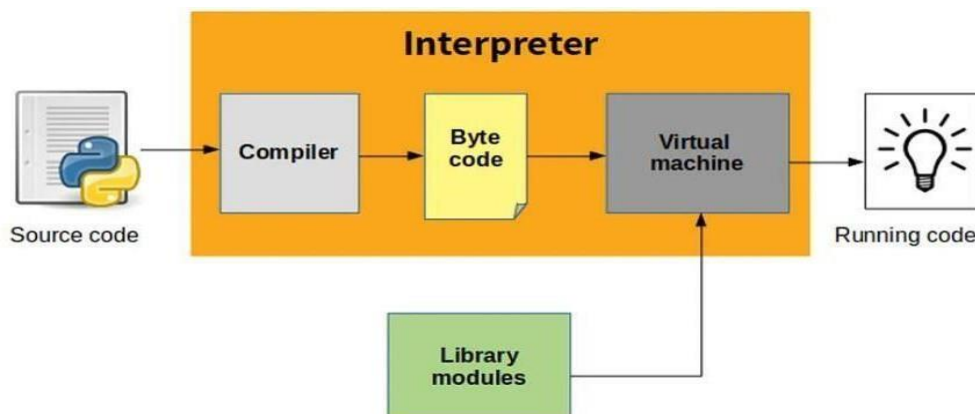
Step 2: After writing Python code it is then saved as a .py file in our system. In this, there are instructions written by a Python script for the system.

Step 3: In this the compilation stage comes in which source code is converted into a byte code. Python compiler also checks the syntax error in this step and generates a .pyc file.

Step 4: Byte code that is .pyc file is then sent to the Python Virtual Machine(PVM) which is the Python interpreter. PVM converts the Python byte code into machine-executable code and in this interpreter reads and executes the given file line by line. If an error occurs during this interpretation, then the conversion is halted with an error message.

Step 5: Within the PVM the byte code is converted into machine code that is the binary language consisting of 0's and 1's. This binary language is only understandable by the CPU of the system as it is highly optimized for the machine code.

Step 6: In the last step, the final execution occurs where the CPU executes the machine code and the final desired output will come as according to your program.



Python was designed for readability, and has some similarities to the English language with influence from mathematics.

- Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.
- Python relies on indentation, using white space, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose.

The Python Platform

- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

MYSQL:

INTRODUCTION:

Database systems play a crucial role in modern digital applications, especially in the domain of digital commerce where large volumes of structured data must be stored, processed, and retrieved efficiently. Among the various relational database management systems (RDBMS), **MySQL** has emerged as one of the most widely used open-source database systems.

MySQL is a relational database management system developed by Oracle Corporation. It is known for its high performance, reliability, scalability, and ease of use. MySQL is widely used in web applications, e-commerce platforms, and enterprise-level systems to manage customer data, product catalogs, transactions, and analytics.

The purpose of this literature survey is to review existing studies, technologies, and methodologies related to MySQL and its applications in digital commerce systems.

Workforce analytics, also known as HR analytics or talent analytics, has emerged as a critical field that combines data analysis, statistical techniques, and computational tools to improve organizational decision-making. Over the past decade, there has been significant growth in research focusing on how employee-related data can be analyzed to understand workforce behavior, predict trends, and enhance organizational performance.

1. Evolution of Workforce Analytics

Employee satisfaction and engagement Early human resource management (HRM) practices relied mainly on intuition and manual record-keeping. However, with the rise of digital systems and big data technologies, organizations started collecting vast amounts of employee data, including demographics, performance metrics, job roles, compensation, and satisfaction levels.

According to a comprehensive literature review on HR analytics, workforce analytics is defined as the use of **research methods and statistical tools to analyze HR data for better decision-making**. It plays a vital role in improving employee performance, predicting workforce trends, and aligning HR strategies with business goals .

Modern work force analytics systems focus on:

- Employee retention and attrition analysis
- Performance evaluation
- Work force planning and forecasting

These systems help organizations transition from reactive HR practices to **data-driven strategic management**.

2. Role of Data Analytics in Workforce Studies

With the growth of data science, workforce analytics has become a multidisciplinary domain involving statistics, machine learning, and database systems. Studies in labor market analytics highlight that workforce data analysis typically follows three main steps:

1. Data collection
2. Data preprocessing(cleaning and transformation)
3. Application of analytical techniques for insights

This structured approach is widely adopted in workforce-related research to ensure accuracy and reliability of results.

Additionally, workforce analytics helps organizations:

- Identify high-performing employees
- Detect patterns of employee turn over
- Understand relationships between satisfaction and productivity
- Optimize recruitment and training strategies

3. Python in Workforce and Data Analytics

Python has become one of the most widely used programming languages in data analytics due to its simplicity, flexibility, and extensive library ecosystem.

A systematic literature review highlights that Python supports all stages of data analysis, including:

- Data collection
- Data cleaning
- Data manipulation
- Data visualization

Popular Python libraries used in workforce analytics include:

- **NumPy** – numerical computations
- **Pandas** – data manipulation and analysis
- **Matplotlib/Seaborn** – visualization
- **Scikit-learn** – machine learning models

Python enables organizations to process large workforce datasets efficiently and generate insights such as:

- Attrition prediction models
- Performance classification
- Salary and experience correlation analysis

Research also shows that Python-based HR analytics systems can significantly improve employee performance tracking and organizational efficiency by integrating machine learning techniques.

4. My SQL in Workforce Data Management

My SQL is a widely used relational database management system(RDBMS) that plays a crucial role in storing and managing workforce data.

In work force analytics systems:

- Employee data is stored in structured tables
- Relationships between data(e.g. employee–department–salary) are maintained
- Queries are used to retrieve and analyze data efficiently

A study on integrating MySQL with analytics systems demonstrates how organizations can build **data warehouses using MySQL** and perform advanced analysis using Python. The system uses structured schemas (such as star schema) to organize data and support reporting and predictive analytics .

Advantages of My SQL in work force analytics:

- Efficient storage of large employee datasets
- Support for complex queries
- Integration with Python through connectors(e.g. My SQL Connector, SQLAlchemy)
- High reliability and scalability

5. Integration of Python and MySQL

The combination of Python and My SQL has become a standard approach in workforce analytics systems.

Typical work flow:

1. Data is stored in My SQL databases
2. Python retrieves data using SQL queries
3. Data is processed and analyzed using Python libraries
4. Results are visualized and used for decision-making

This integration allows organizations to:

- Automate data analysis processes
- Perform real-time work force monitoring
- Build predictive models for attrition and performance

Research indicates that combining database systems with analytical tools enhances decision-making capabilities and improves organizational efficiency.

6. Applications of Workforce Analytics

Workforce analytics has been applied across various domains to improve organizational outcomes. Key applications include:

a) Employee Attrition Analysis

Organizations analyze historical employee data to identify factors causing turnover, such as:

- Low job satisfaction
- Poor compensation
- Lack of growth opportunities

b) Performance Analysis

Data-driven models help evaluate employee performance based on:

- Experience
- Department
- Work environment

c) Work force Planning

Analytics enables forecasting of future work force needs, helping organizations:

- Plan recruitment strategies
- Allocate resources effectively

d) Employee Satisfaction Analysis

By analyzing survey data and workplace conditions, organizations can identify factors affecting employee morale and productivity.

HR analytics research shows that these applications contribute to:

- Improved ROI
- Better work force utilization
- Enhanced strategic decision-making

7. Use of Machine Learning in Workforce Analytics

Recent studies highlight the growing use of machine learning techniques in work force analytics.

These techniques help in:

- Predicting employee attrition
- Classifying employee performance levels
- Identifying hidden patterns in workforce data

Artificial intelligence and data science have significantly enhanced talent analytics by enabling organizations to make **real-time and predictive decisions** about workforce management .

8. Challenges in Workforce Analytics

Despite its advantages, work force analytics faces several challenges:

- **Data Quality Issues**—incomplete or inconsistent employee data
- **Privacy Concerns**—handling sensitive employee information
- **Integration Complexity**—combining multiple data sources
- **Skill Gap**—lack of expertise in data analytics tools
- **Interpretation Issues**—difficulty in translating insights into actions

Studies emphasize that organizations need proper infrastructure ,skilled professionals, and ethical frameworks to effectively implement workforce analytics.

WORKFORCE ANALYTICS – PYTHON & MYSQL

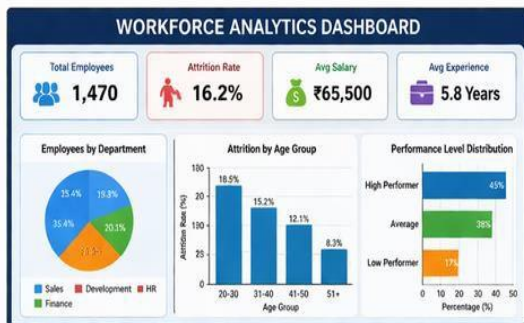


Figure 1: Workforce Analytics Dashboard

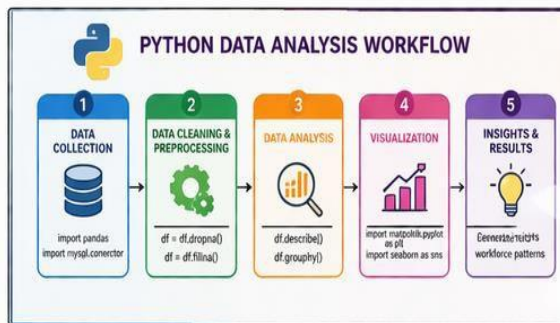


Figure 2: Python Data Analysis Workflow

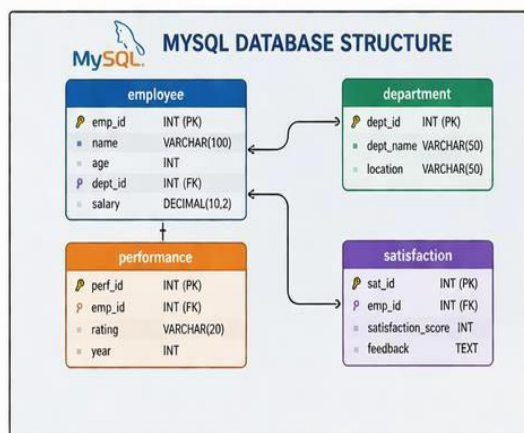


Figure 3: MySQL Database Structure for Employee Data

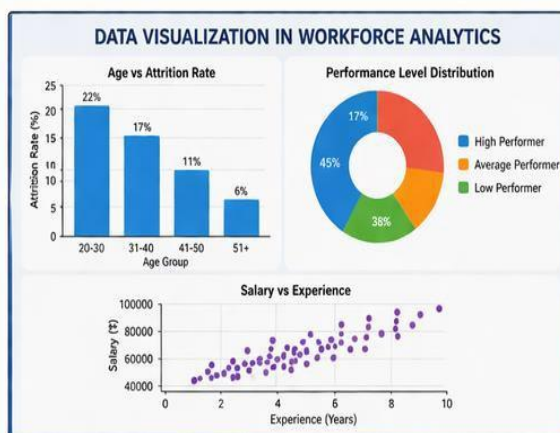


Figure 4: Data Visualization of Employee Patterns

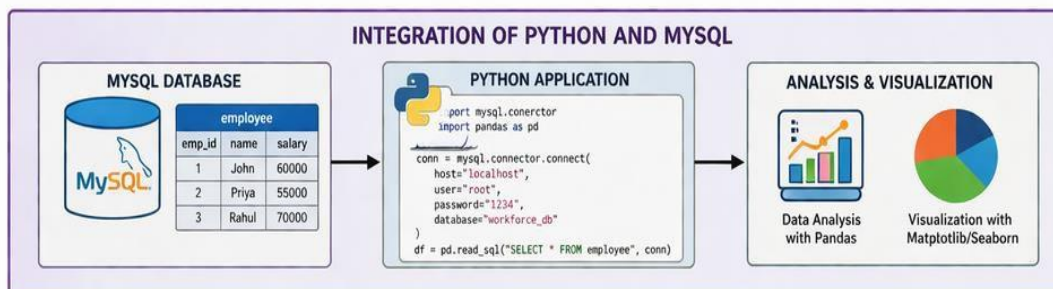


Figure 5: Integration of Python with MySQL for Workforce Analytics

3. SYSTEM ANALYSIS

Feasibility Study:

The feasibility study of this project focuses on evaluating whether the workforce analytics system using Python and MySQL can be successfully implemented in terms of technical, economic, and operational aspects. This project is technically feasible because it uses widely available technologies such as Python and MySQL, which are easy to learn and implement. Python provides powerful libraries for data analysis and visualization, while MySQL ensures efficient storage and management of employee data. The required tools and software can be easily installed on standard computer systems without the need for advanced hardware, making the system simple to develop and execute.

From an economic perspective, the project is cost-effective as it uses open-source technologies. Python and MySQL are freely available, which reduces the overall development cost. There is no need for expensive software or licenses, making it suitable for small and medium-level projects. The system can be developed with minimal resources while still providing accurate and useful results. This makes the project economically feasible and accessible for organizations or academic purposes.

The operational feasibility of the system is also high, as it is user-friendly and easy to understand. The process of storing data in MySQL and analyzing it using Python is straightforward and does not require complex operations. The system can be used by individuals with basic knowledge of databases and programming. The results generated through analysis and visualization are easy to interpret, which helps users in making better decisions. This ensures that the system can be effectively used in real-world scenarios.

In addition, the project is feasible in term soft time, as it can be completed within a limited period using available tools and datasets. The steps involved, such as data collection, storage, analysis, and visualization, can be implemented in a systematic manner without requiring excessive time. This makes the project suitable for academic submission within a given deadline.

Overall, the feasibility study indicates that the workforce analytics system using Python and MySQL is practical, cost-effective, and easy to implement. It can be successfully developed and used to analyze employee data, identify patterns, and support better workforce management decisions.

Existing System:

The existing system is primarily used for storing and viewing employee related data in a structured format. Information such as employee age, department, job role, salary, experience, and other details is maintained in tabular form, usually through spreadsheets or basic database systems. This allows users to record and access employee information whenever required.

The system enables simple operations like adding new records, updating existing data, and retrieving information for reference. Data is generally organized in rows and columns, making it easy to view individual employee details and perform basic sorting or filtering.

In the existing system, workforce management is mainly handled using traditional methods such as manual record-keeping and basic software tools like spreadsheets. Organizations store employee data in separate files or simple database systems without proper integration or advanced analysis. The data related to employee demographics, job roles, salary, performance, and attrition is maintained, but it is not effectively analyzed to identify patterns or trends. Decision-making is mostly based on past experience and assumptions rather than data-driven insights.

In many cases, organizations use basic reporting techniques to understand employee information, which provides only limited insights. There is no proper use of advanced tool sort technologies to analyze large volumes of workforce data. As a result, it becomes difficult to identify important factors affecting employee performance, satisfaction, and retention. The existing system lacks automation, making the process time-consuming and less efficient.

Additionally, the absence of integration between data storage and analysis tools makes it harder to process data effectively. The system does not support real-time analysis or visualization, which limits the ability of management to make quick and informed decisions.

Draw backs of the Existing System:

1. Time-Consuming Process

- Manual data entry and maintenance takes more time
- Data processing is slow
- Report generation is delayed

2. Lack of Data Analysis

- No advanced tools for analyzing workforce data
- Difficult to identify patterns and trends
- Limited insights from available data

3. Poor Decision Making

- Decisions are based on assumptions
- Lack of data-driven strategies
- High chances of incorrect decisions

4. No Real-Time Analysis

- Data is not updated instantly
- Cannot monitor employee performance in realtime
- Delay in identifying issues

5. Difficulty in Handling Large Data

- Inefficient when data size increases
- Storage and retrieval become complex
- System performance decreases

6. Lack of Data Visualization

- No proper graphs or charts
- Difficult to understand complex data
- Poor presentation of results

7. No Integration Between Systems

- Data stored in separate files or systems
- No connection between database and analysis tools
- Increased manual work

8. High Risk of Errors

- Manual handling leads to human errors
- Data in consistency issues
- Incorrect reports may be generated

9. Poor Attrition Analysis

- Cannot identify reasons for employee turn over
- No prediction of employee leaving
- Difficulty in improving retention

10. Limited Support for Planning

- No proper work force forecasting
- Difficulty in resource allocation
- Ineffective future planning

DISADVANTAGES OF EXISTING SYSTEM:

1. Slow Processing of Data

The existing system depends heavily on manual data handling, which makes the overall process slow and inefficient. Data entry, updating, and report generation take a lot of time, especially when dealing with large amounts of employee information. This delay affects timely decision-making and reduces productivity.

2. Limited Analytical Capability

The system does not support advanced data analysis techniques, making it difficult to identify patterns and relationships in workforce data. It only provides basic information without deeper insights, which limits the ability of organizations to understand employee behavior effectively.

3. Inaccurate Decision Making

Since the existing system lacks proper data analysis, most decisions are based on assumptions rather than actual data. This increases the chances of incorrect conclusions and poor decision-making, which can negatively impact organizational performance.

4. No Real-Time Monitoring

The system does not provide real-time data updates, which makes it difficult to track employee performance and other activities instantly. As a result, there is a delay in identifying problems and taking necessary actions, affecting overall efficiency.

5. Poor Data Management

In the existing system, data is often stored in an unorganized manner, making it difficult to manage and retrieve information. This can lead to data loss, duplication, and inconsistency, which reduces the reliability of the system.

6. Lack of Visualization Tools

The system does not include proper visualization features such as graphs and charts. This makes it difficult to understand complex data and present it in a clear and meaningful way, which limits effective communication of insights.

7. High Dependency on Manual Work

The existing system relies heavily on manual processes, which increases workload and reduces efficiency. Employees need to spend more time handling data, which can lead to fatigue and reduced productivity.

8. Error-Prone System

Manual data handling increases the chances of human errors such as incorrect data entry and calculations. These errors can affect the accuracy of results and lead to wrong conclusions.

9. No Predictive Analysis

The system does not support predictive techniques, making it difficult to forecast future workforce trends. Organizations cannot predict employee attrition or plan future work force needs effectively.

10. Not Scalable

The existing system is not suitable for handling large datasets or growing organizational needs. As the amount of data increases, the system becomes less efficient and difficult to manage.

Proposed System:

The proposed system is designed to transform basic employee data into structured, meaningful insights through a well defined data analytics workflow. Instead of only storing and viewing records, this system focuses on organizing, processing, analyzing, and visualizing workforce data in a systematic manner. It integrates multiple technologies to ensure that data is clean, consistent, and ready for analysis, thereby improving efficiency and supporting better organizational planning.

The system begins with raw employee data collected in the form of Excel or CSV files. This data includes attributes such as employee ID, age, department, job role, salary, experience, performance rating, satisfaction level, attrition status, and other related fields. Since raw data may contain inconsistencies, missing values, or duplicate records, the first step of the system focuses on data cleaning and preparation.

Using pandas and numpy in a Jupyter Notebook environment, the dataset is processed to ensure quality and consistency. Missing values are handled appropriately, duplicate entries are removed, and incorrect formats are corrected. Columns are standardized, and new derived columns can be created where necessary to support analysis. This step ensures that the dataset becomes reliable and structured for further operations.

Once the data is cleaned, it is transformed into a structured format and stored in a SQL database. This step plays a crucial role in improving data management. SQL provides a centralized system where all employee data is stored in tables, allowing efficient access and manipulation. Structured queries are used to extract meaningful information from the dataset. For example, queries can calculate total employees, count attrition cases, group employees by department, and analyze salary distribution. This approach significantly improves the speed and accuracy of data retrieval compared to manual methods.

The core logic of the proposed system lies in using SQL queries for detailed data analysis. Instead of manually filtering and calculating values, SQL enables operations such as grouping, aggregation, sorting, and conditional filtering. This allows users to analyze workforce patterns such as which departments have higher attrition, how salary varies across roles, and how

employee satisfaction differs among groups. By executing these queries, the system converts raw data into structured insights that are easy to interpret.

In addition to SQL based analysis, matplotlib is used within Jupyter Notebook to create visual representations of the data. Charts such as bar graphs, line charts, and distribution plots help in identifying patterns and trends more clearly. For instance, visualizing attrition across departments or salary distribution across job roles provides a better understanding compared to raw tables. These visualizations act as an intermediate step before final reporting.

The final stage of the system involves connecting the processed data to Power BI for dashboard creation. The cleaned and structured dataset, either in Excel or CSV format, is imported into Power BI. Using this data, interactive dashboards are created with cards and charts. Cards display key metrics such as total employees, attrition count, average salary, and employee distribution. Charts represent relationships between variables such as department and attrition, job role and salary, and satisfaction levels across different employee groups. This visual layer makes it easier for users to explore and understand workforce data.

The workflow of the proposed system can be summarized in a sequence of steps. First, raw data is collected and loaded into Jupyter Notebook. Second, data cleaning and preprocessing are performed using pandas and numpy. Third, the cleaned data is stored in a SQL database for structured querying. Fourth, SQL queries are executed to perform analysis and extract insights. Fifth, matplotlib is used to create initial visualizations. Finally, the processed dataset is connected to Power BI to build dashboards for reporting and presentation.

This system introduces a clear improvement over the existing system by moving from simple data storage to structured data analysis. Data is no longer just recorded but is actively processed and analyzed to extract meaningful patterns. The use of SQL ensures faster and more efficient querying, while visualization tools improve data interpretation. The integration of multiple technologies creates a seamless pipeline from raw data to final insights.

Another key improvement is the reduction of manual effort. Tasks such as filtering data, calculating metrics, and generating reports are automated through code and queries. This reduces

the chances of human error and ensures consistency in results. Data validation and cleaning steps further improve accuracy, making the dataset more reliable for analysis.

The proposed system also supports better scalability. As the volume of data increases, SQL databases can handle large datasets more efficiently compared to basic storage methods. Queries can be executed quickly even on large tables, ensuring that performance is maintained. This makes the system suitable for growing organizations with increasing workforce data.

In terms of data accessibility, the centralized SQL database and PowerBI dashboards allow users to access information easily. Instead of searching through multiple files, users can retrieve required data through queries or view summarized insights through dashboards. This improves collaboration and ensures that all users work with consistent data.

The logic of the system is based on transforming raw data into actionable insights through a step by step process. Each stage adds value to the data. Cleaning ensures accuracy, SQL enables structured analysis, visualization highlights patterns, and dashboards present the final results in an understandable format. This logical flow ensures that the output is both reliable and meaningful.

Overall, the proposed system provides a comprehensive solution for workforce analysis. It enhances data handling, improves efficiency, and enables deeper understanding of employee related patterns. By combining data processing, querying, and visualization, the system supports better organizational planning and decision making without relying on complex methods.

Advantages of the Proposed System:

1. Faster Data Processing

The proposed system uses Python and MySQL, which significantly improve the speed of data processing compared to manual methods. Data can be stored, retrieved, and analyzed quickly, allowing faster report generation and decision-making. This reduces delays and increases overall efficiency.

2. Advanced Data Analysis

The system supports advanced data analysis using Python libraries, enabling deeper insights into workforce data. It helps in identifying patterns, relationships, and trends related to employee performance, satisfaction, and attrition. This makes the analysis more accurate and meaningful.

3. Data-Driven Decision Making

Unlike the existing system, the proposed system is based on data-driven approaches. Decisions are made using actual data and analysis results, which improves accuracy and reduces the chances of errors. This leads to better organizational planning and management.

4. Real-Time Data Handling

The proposed system allows faster data updates and processing, making it easier to monitor employee data in near real-time. This helps organizations quickly identify issues and take immediate actions to improve performance and efficiency.

5. Efficient Data Management

Using MySQL, the system stores data in a structured and organized format. This ensures easy access, better data consistency, and reduced chances of data loss. It also supports handling large amounts of workforce data effectively.

6. Better Data Visualization

The system uses visualization tools like Matplotlib and Seaborn to present data in the form of graphs and charts. This makes it easier to understand complex data and communicate insights clearly to management.

7. Reduced Manual Work

Automation in data processing and analysis reduces the need for manual work. This minimizes human effort, saves time, and increases productivity. It also reduces the chances of errors caused by manual handling.

8. Improved Accuracy

Since the system uses automated tools and structured data processing, it provides more accurate results. This improves the reliability of analysis and helps in making correct decisions.

9. Predictive Analysis Capability

The proposed system can be extended to include predictive analysis using machine learning techniques. This helps in forecasting employee attrition and workforce needs, allowing organizations to plan better for the future.

10. Scalable System

The system is designed to handle large datasets and can be easily scaled as the organization grows. It can manage increasing amounts of data without affecting performance, making it suitable for long-term use.

Hardware Requirements:

- Computer or Laptop
- Minimum 4GBRAM
- Intel i3 Processor or above
- 256GB Hard Disk/ SSD
- Keyboard and Mouse
- Stable Power Supply

Software Requirements:

- Python Programming Language
- My SQL Database
- Python IDE (Jupyter Notebook /VS Code/PyCharm)
- Pandas Library
- NumPy Library
- Matplotlib Library
- Seaborn Library
- MySQL Workbench/SQL Client
- Operating System(Windows/Linux)

4. SYSTEM DESIGN

Modules of the System:

1. DataCollection Module

Purpose:

- To gather employee-related data for analysis

Features:

- Collects data such as age, job role, salary, experience
- Supports dataset input from files or records
- Prepares raw data for further processing

2. Data Storage Module

Purpose:

- To store and manage work force data efficiently

Features:

- Uses My SQL data base for storage
- Organizes data in to tables
- Ensures secure and structured data management

3. Data Preprocessing Module

Purpose:

- To clean and prepare data for analysis

Features:

- Handles missing values
- Removes duplicate records
- Converts data into suitable format

4. Data Analysis Module**Purpose:**

- To analyze workforce data and identify patterns

Features:

- Uses Python libraries like Pandas and NumPy
- Performs filtering, grouping, and aggregation
- Identifies trends and relationships

5. Data Visualization Module**Purpose:**

- To represent data in graphical form

Features:

- Generates charts and graphs
- Uses Matplotlib and Seaborn
- Improves understanding of data

6. Attrition Analysis Module**Purpose:**

- To study employee turn over patterns

Features:

- Identifies reasons for attrition
- Detects high-risk employee groups
- Helps improve retention strategies

7. Performance Analysis Module**Purpose:**

- To evaluate employee performance

Features:

- Compares performance across departments
- Identifies high and low performers
- Supports performance improvement

8. Reporting Module**Purpose:**

- To generate final results and insights

Features:

- Creates summary reports
- Presents analysis results clearly
- Helps in decision-making

9. User Interface Module**Purpose:**

- To provide easy interaction with the system

Features:

- Simple and user-friendly interface
- Allows users to view reports and data
- Easy navigation

10. Database Connectivity Module**Purpose:**

- To connect Python with My SQL database

Features:

- Retrieves data from database
- Sends processed data if needed
- Ensures smooth data communication

Block Diagram:

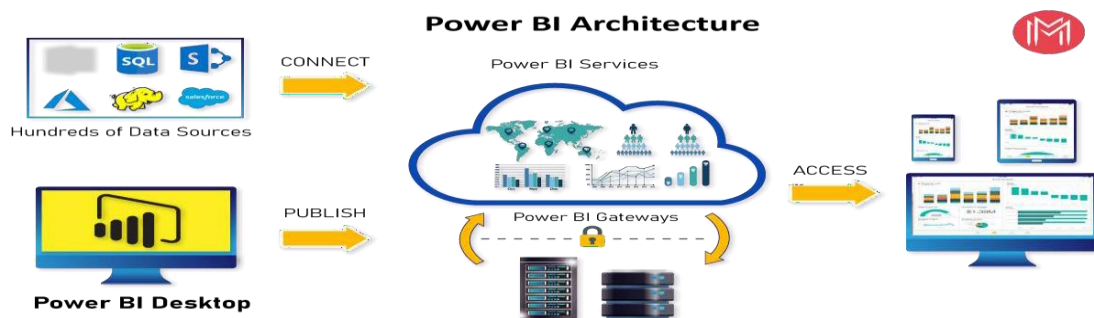
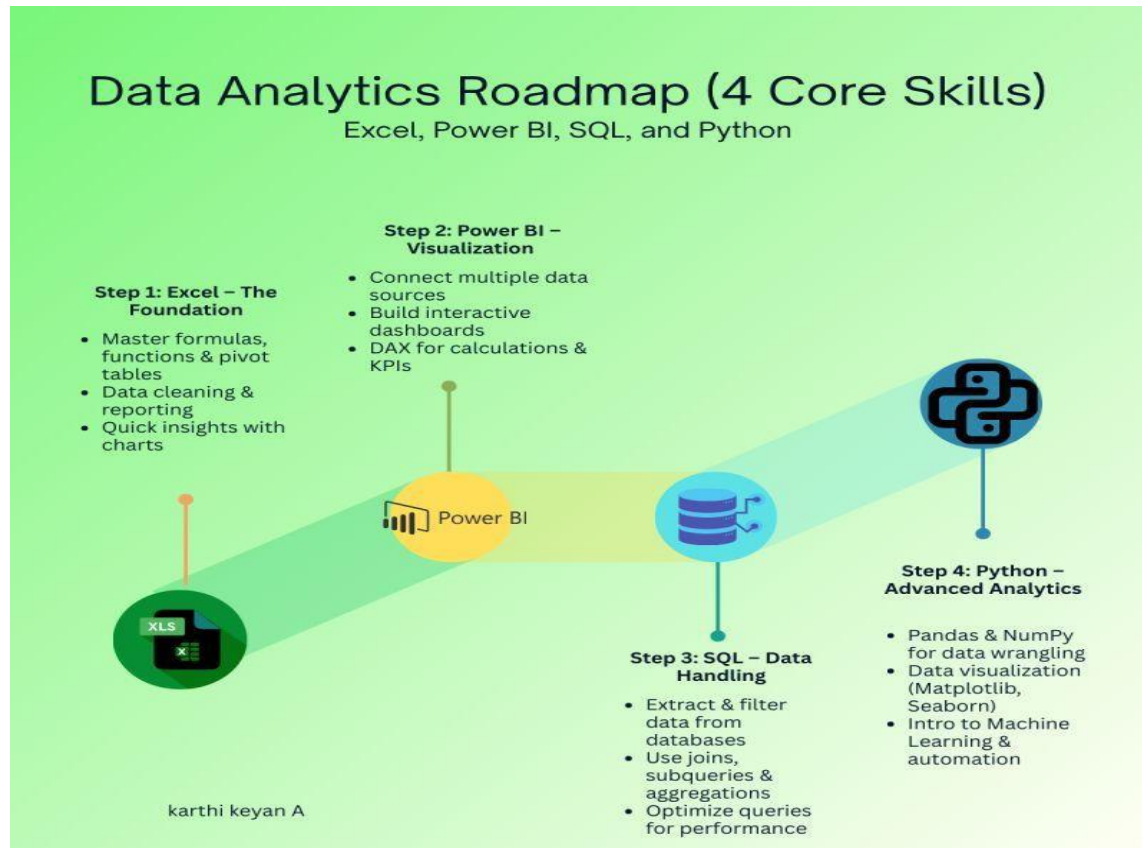
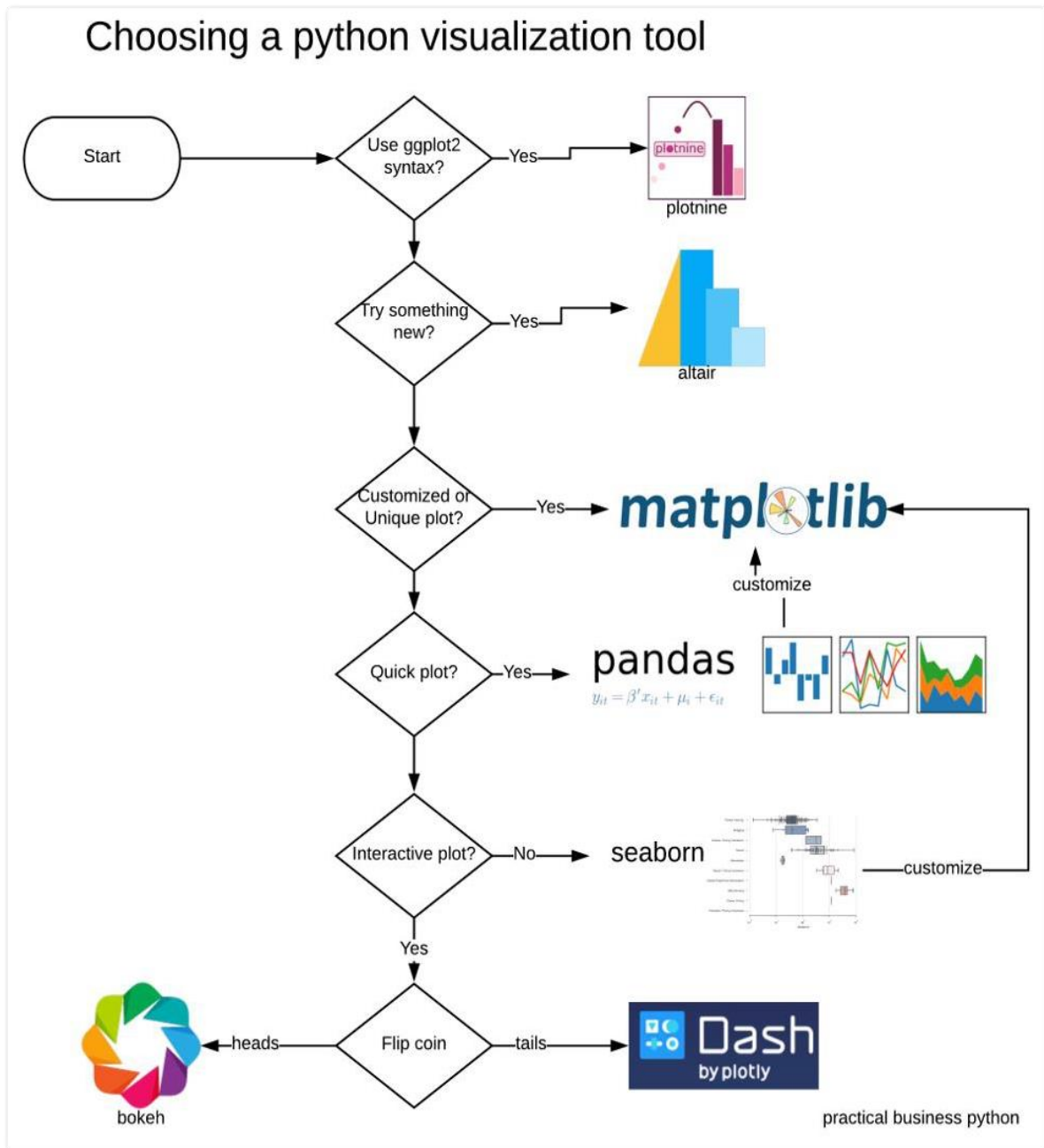


FIGURE 4.1: Block Diagram For User



**FIGURE 4.2: Block Diagram For Python
Visualization Tool**

Explanation of Each Block:

Data Source:

This is the starting point where raw employee data is available. The data is stored in Excel or CSV files and includes details such as employee ID, age, department, salary, job role, performance, and attrition. At this stage, the data may not be clean and can contain missing values or inconsistencies.

SQL Database:

In this stage, the data is stored in a structured format using a SQL database. Tables are created to organize employee data properly. This block handles storing large datasets and allows easy access to data using queries. It ensures that data is centralized and well structured.

Python Processing:

Here, Python is used in Jupyter Notebook with pandas and numpy. The data is cleaned, formatted, and prepared for analysis. Operations like removing duplicates, handling missing values, and correcting data types are performed. SQL queries are also used to analyze the data and extract useful insights.

Visualization (PowerBI):

The processed data is connected to Power BI. In this stage, charts, graphs, and cards are created to represent the data visually. It helps in understanding patterns such as employee distribution, attrition trends, and salary variations in a simple way.

User:

The final stage is the user who views the dash boards. The user interacts with visual reports and understands insights without needing technical knowledge. The system presents data in an easy and clear format.

Dataflow Diagram:

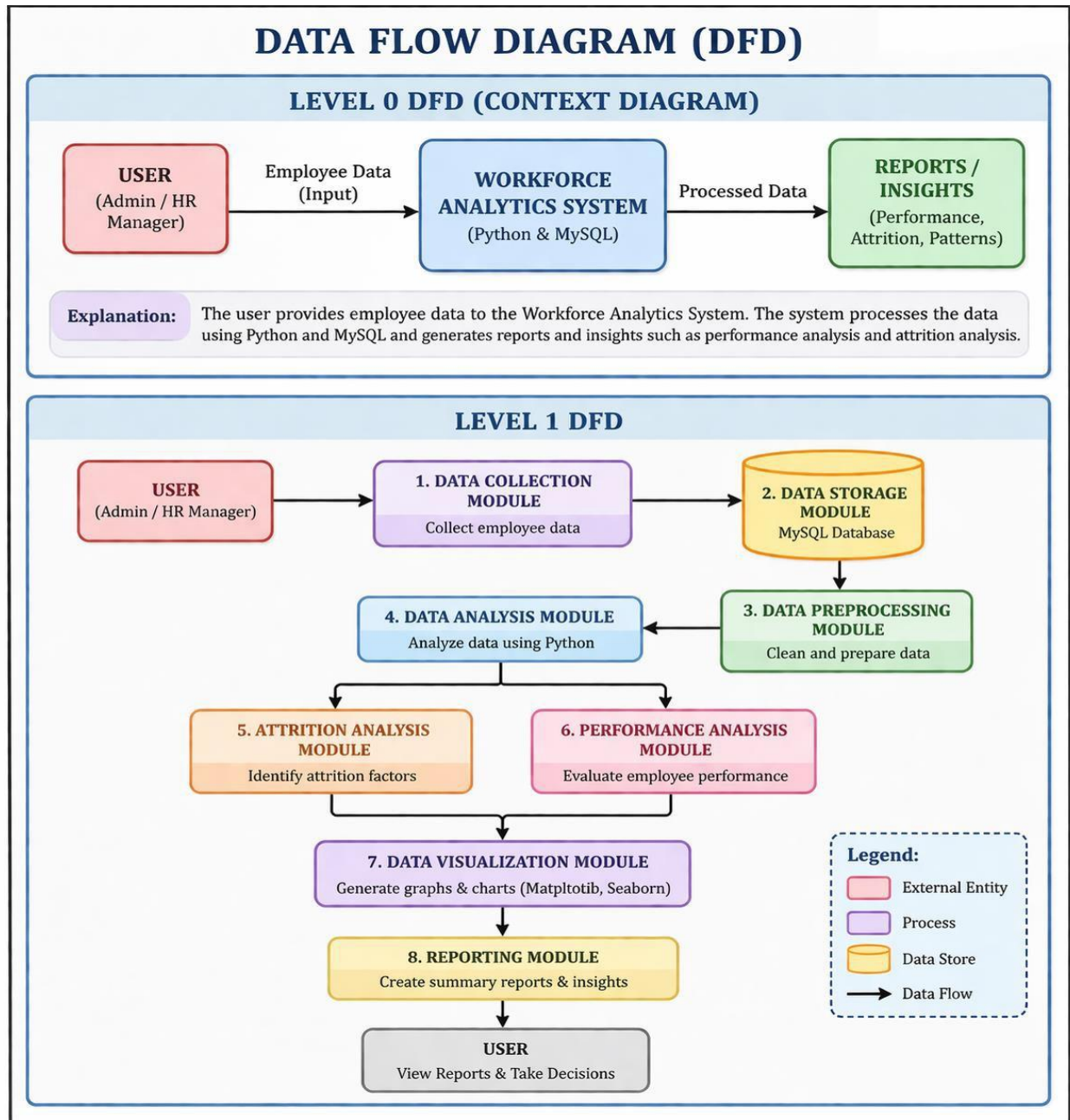


FIGURE 4.3: Data Flow Diagram Of Work Force Analytics System

UML:

The UML diagram represents the structure and functionality of the workforce analytics system. It includes a use case diagram and a class diagram to explain how the system works. The use case diagram shows the interaction between the user and the system, where the user performs actions such as collecting employee data, storing it in the database, preprocessing, analyzing, visualizing, and generating reports. It explains the overall workflow of the system.

The class diagram represents the internal structure of the system by showing different classes such as user, data collector, database, data analyzer, visualizer, and report generator. Each class has its own attributes and methods, and they are connected to show how data flows within the system. The diagram helps in understanding how different components interact with each other to perform workforce analysis.

Overall, the UML diagram provides a clear view of both the functional behavior and structural design of the system, making it easier to understand the implementation process.

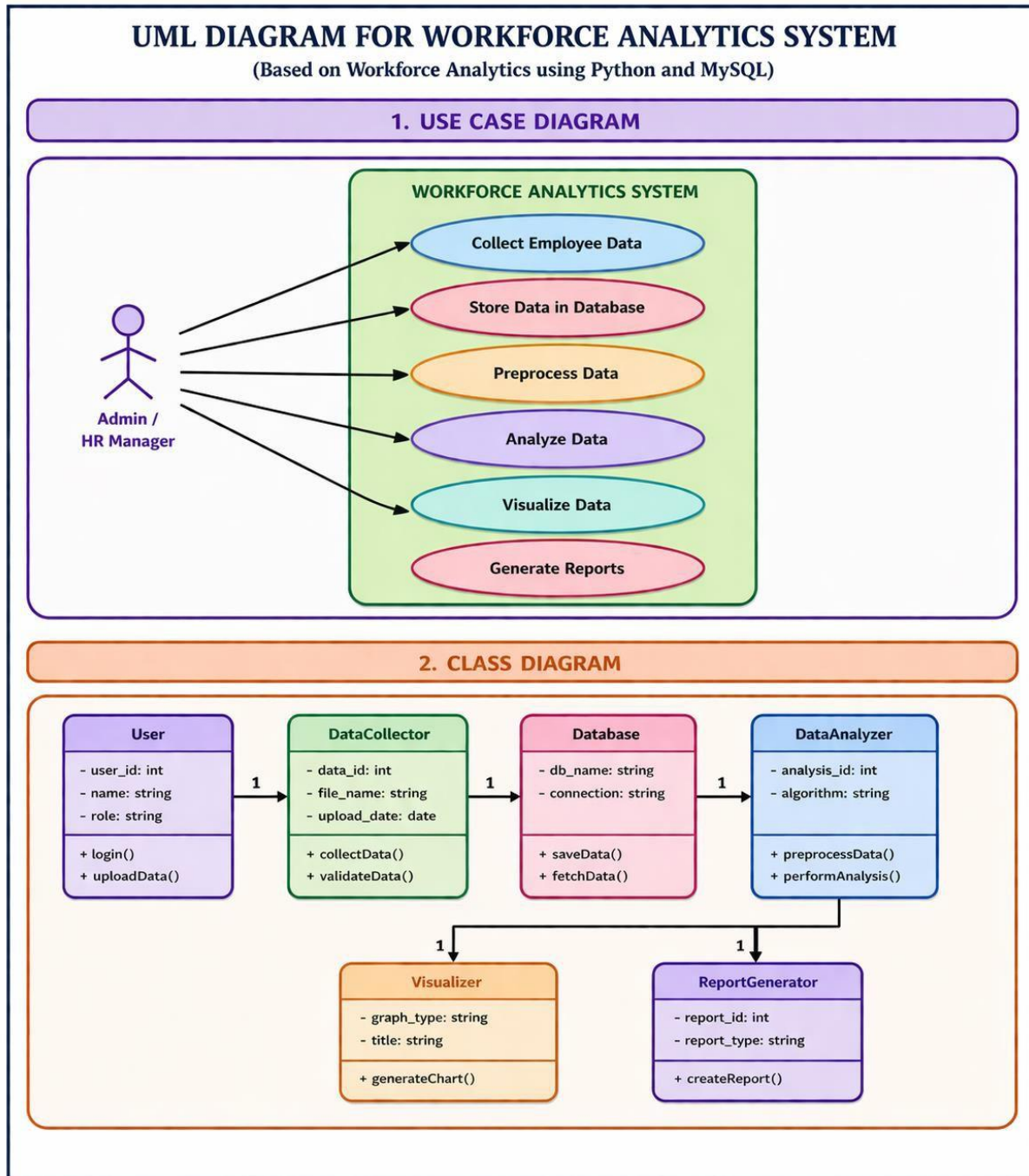


FIGURE 4.4: UML Diagram For Work Force Analytics System

USECASEDIAGRAM:

The use case diagram represents the interaction between the user and the workforce analytics system. It shows how the system functions by illustrating the different activities performed by the user. In this system, the primary user is the admin or HR manager who interacts with the system to manage and analyze employee data. The diagram helps in understanding the overall workflow and how the system responds to user inputs.

The process begins with the user providing employee-related data such as age, job role, salary, experience, performance level, and satisfaction. The system then stores this data in the MySQL database for further processing. After storing the data, the system performs preprocessing to clean and prepare the data by removing errors and handling missing values. This step ensures that the data is accurate and suitable for analysis.

Once the data is prepared, the system performs data analysis using Python to identify patterns and relationships within the workforce data. The system then generates visual representations such as graphs and charts to make the analysis easier to understand. Finally, the system produces reports that provide insights into employee performance, attrition, and overall workforce patterns.

The use case diagram clearly shows the sequence of actions performed by the user and the system, highlighting how data flows from input to output. It helps in understanding the functional requirements of the system and ensures that all necessary operations are properly defined. Overall, the use case diagram provides a clear view of how the workforce analytics system operates and how users interact with it.

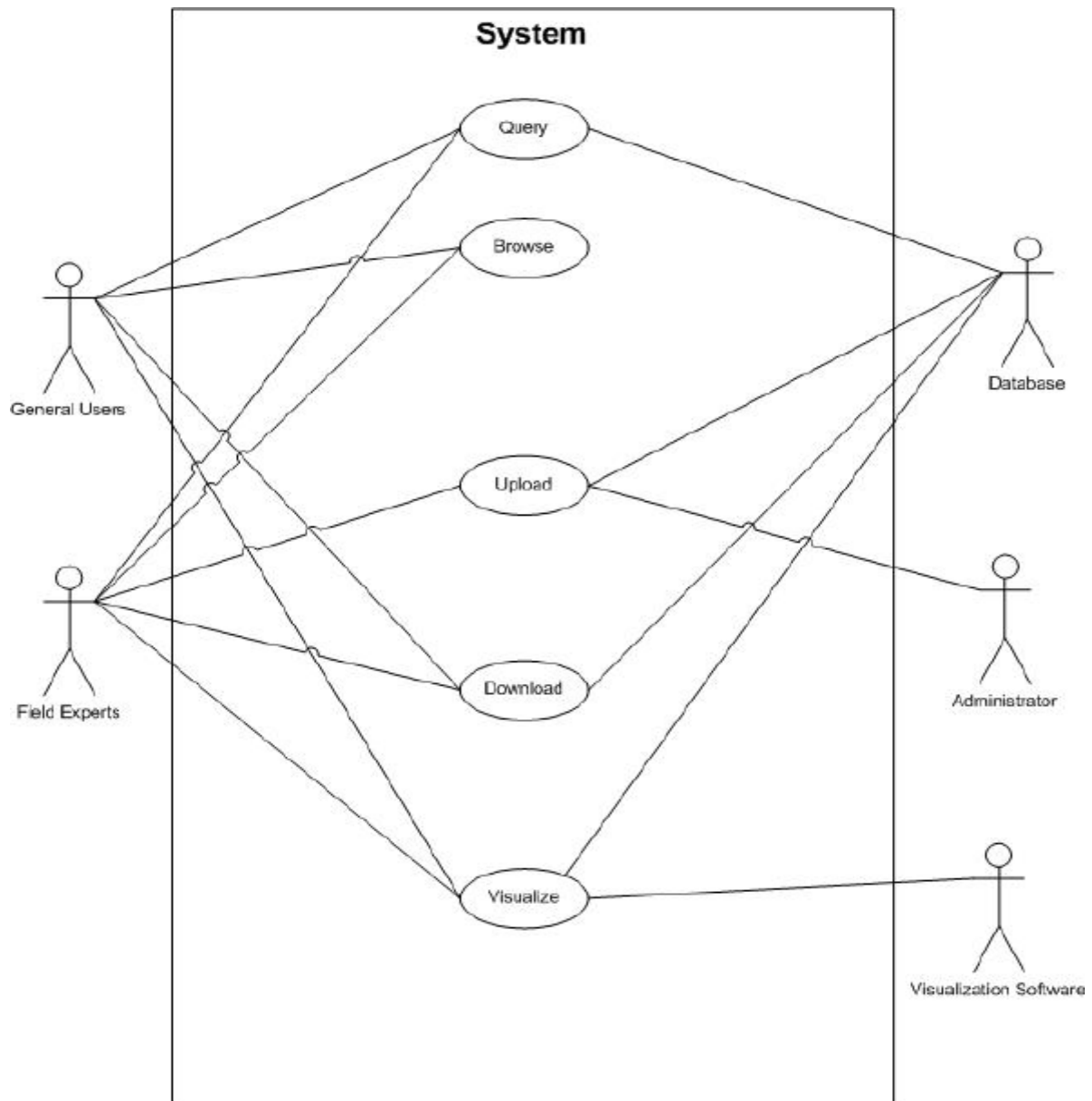


FIGURE 4.5: Use Case Diagram Of System

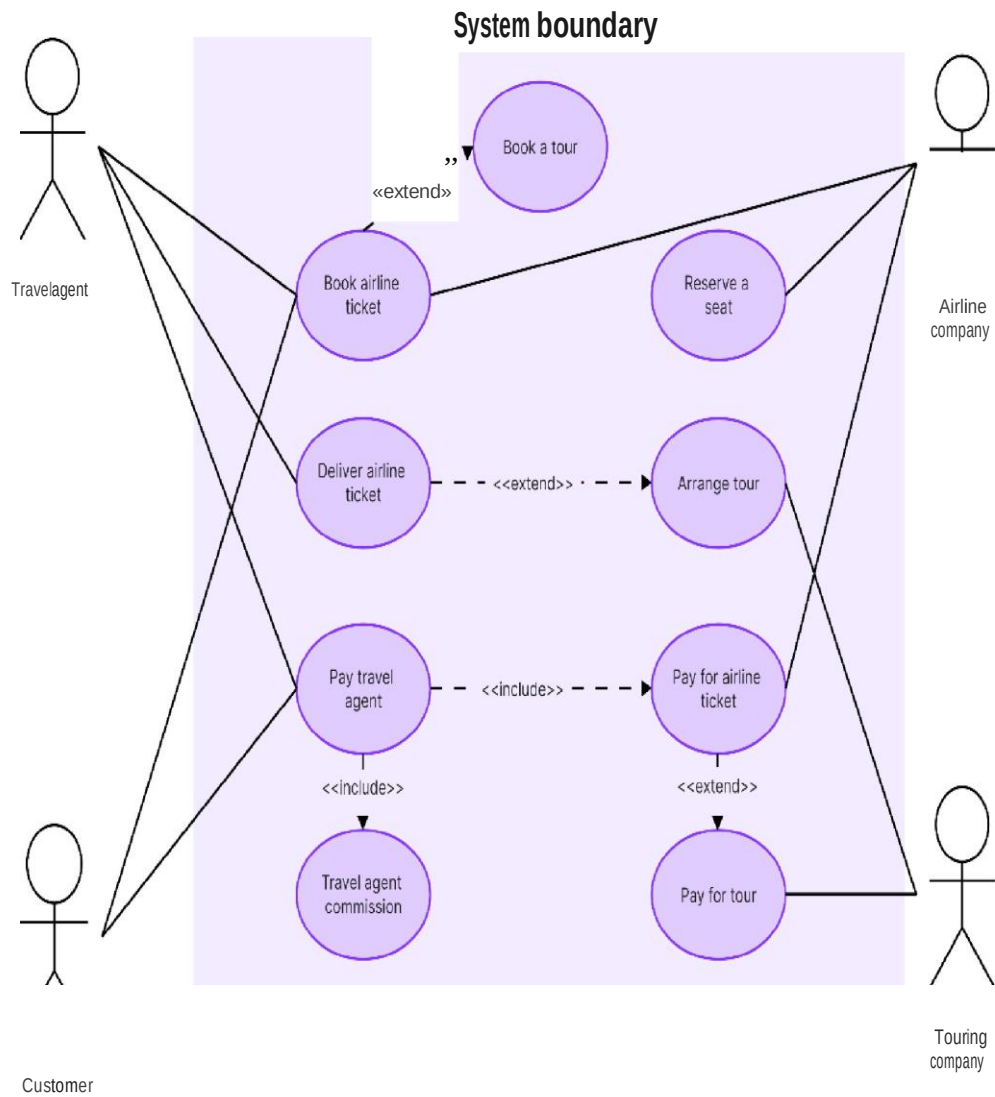


FIGURE 4.6: Use Case Diagram Of System Boundary

SEQUENCEDIAGRAM:

The sequence diagram represents the step-by-step interaction between different components of the workforce analytics system over time. It shows how data flows from the user to the system and how different modules communicate with each other to complete the analysis process. This diagram helps in understanding the order of operations and the sequence in which tasks are performed.

The process begins when the user, such as an admin or HR manager, provides employee-related data to the system. This data is first handled by the data collection module, which gathers and prepares the input for further processing. The collected data is then sent to the database module, where it is stored in the MySQL database in a structured format.

After storing the data, the system retrieves it from the database and passes it to the preprocessing module. In this step, the data is cleaned by removing missing values, duplicates, and inconsistencies to ensure accuracy. Once the data is preprocessed, it is forwarded to the data analysis module, where Python is used to analyze the data and identify patterns, relationships, and trends related to employee performance and attrition.

Following the analysis, the results are sent to the visualization module, which converts the data into graphical representations such as charts and graphs. These visual outputs make it easier for users to understand the analysis results. Finally, the reporting module generates reports based on the analyzed data and presents them to the user.

The sequence diagram clearly illustrates the interaction between the user, data collection, database, preprocessing, analysis, visualization, and reporting modules. It shows how each component works in a specific order to achieve the final output. Overall, the diagram provides a clear understanding of the system's workflow and the communication between different modules during execution.

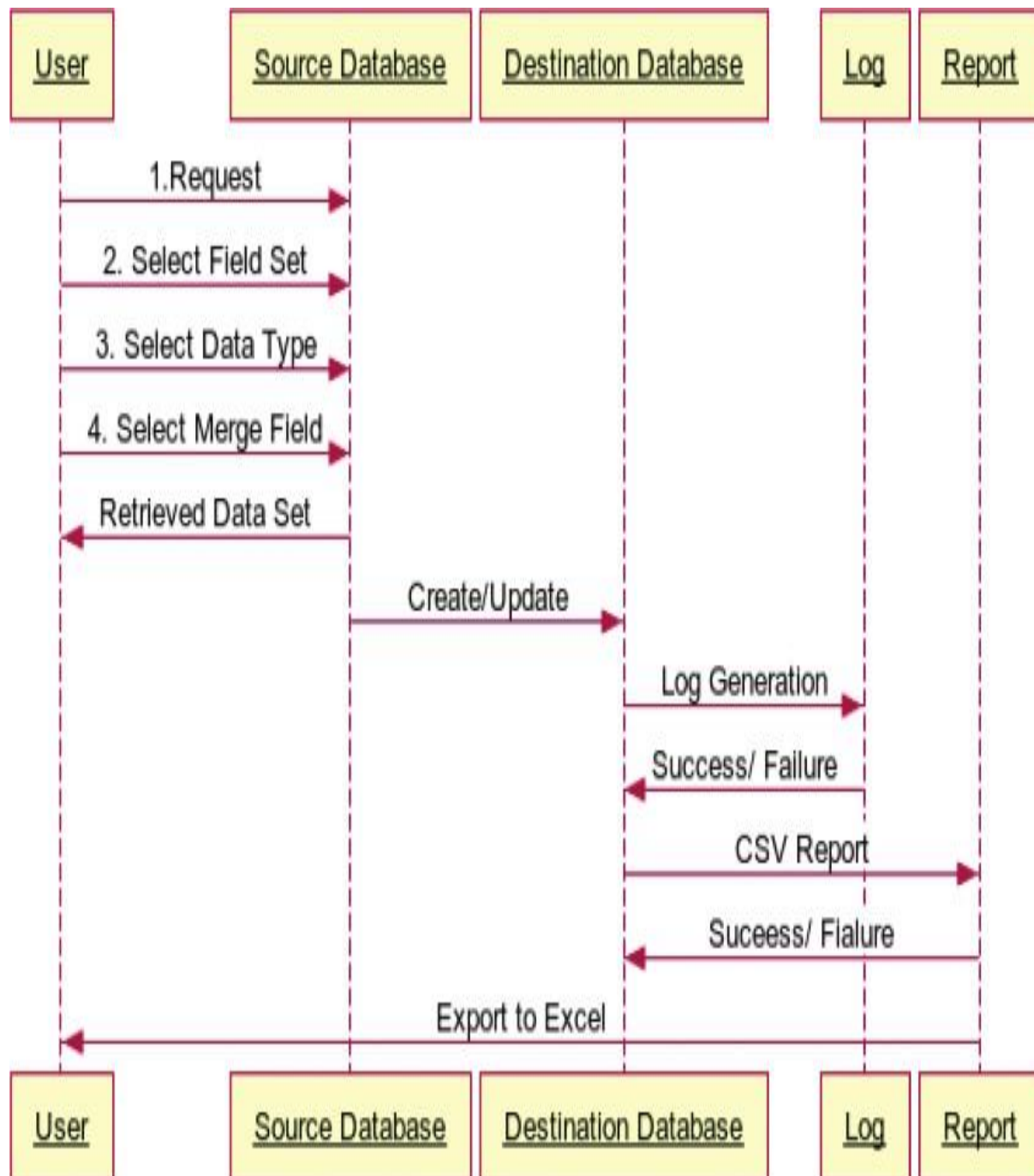


FIGURE 4.7: Sequence Diagram For User

CLASSDIAGRAM:

The class diagram represents the structural design of the workforce analytics system by showing the different classes, their attributes, methods, and relationships. It provides a clear understanding of how the system is organized and how various components interact with each other to perform data analysis. The diagram includes major classes such as user, data collector, database, data analyzer, visualizer, and report generator, which together form the core of the system.

The user class represents the admin or HR manager who interacts with the system. It includes attributes such as user ID, name, and role, and methods like login and data upload. The data collector class is responsible for gathering employee data and validating it before processing. It includes attributes related to the dataset and methods for collecting and verifying data.

The database class handles the storage and retrieval of employee data using MySQL. It ensures that data is stored in a structured format and can be accessed whenever required. The data analyzer class is one of the most important components, as it performs preprocessing and analysis of the data using Python. It identifies patterns, relationships, and trends related to employee performance, satisfaction, and attrition.

The visualizer class is used to convert the analyzed data into graphical representations such as charts and graphs, making it easier to understand the results. The report generator class creates final reports based on the analysis and visualization, providing useful insights for decision-making. These classes are connected through relationships that show how data flows from one component to another within the system.

Overall, the class diagram helps in understanding the internal structure of the workforceanalytics system and how different modules work together. It provides a clear view of the system design, making it easier to implement and maintain the project.

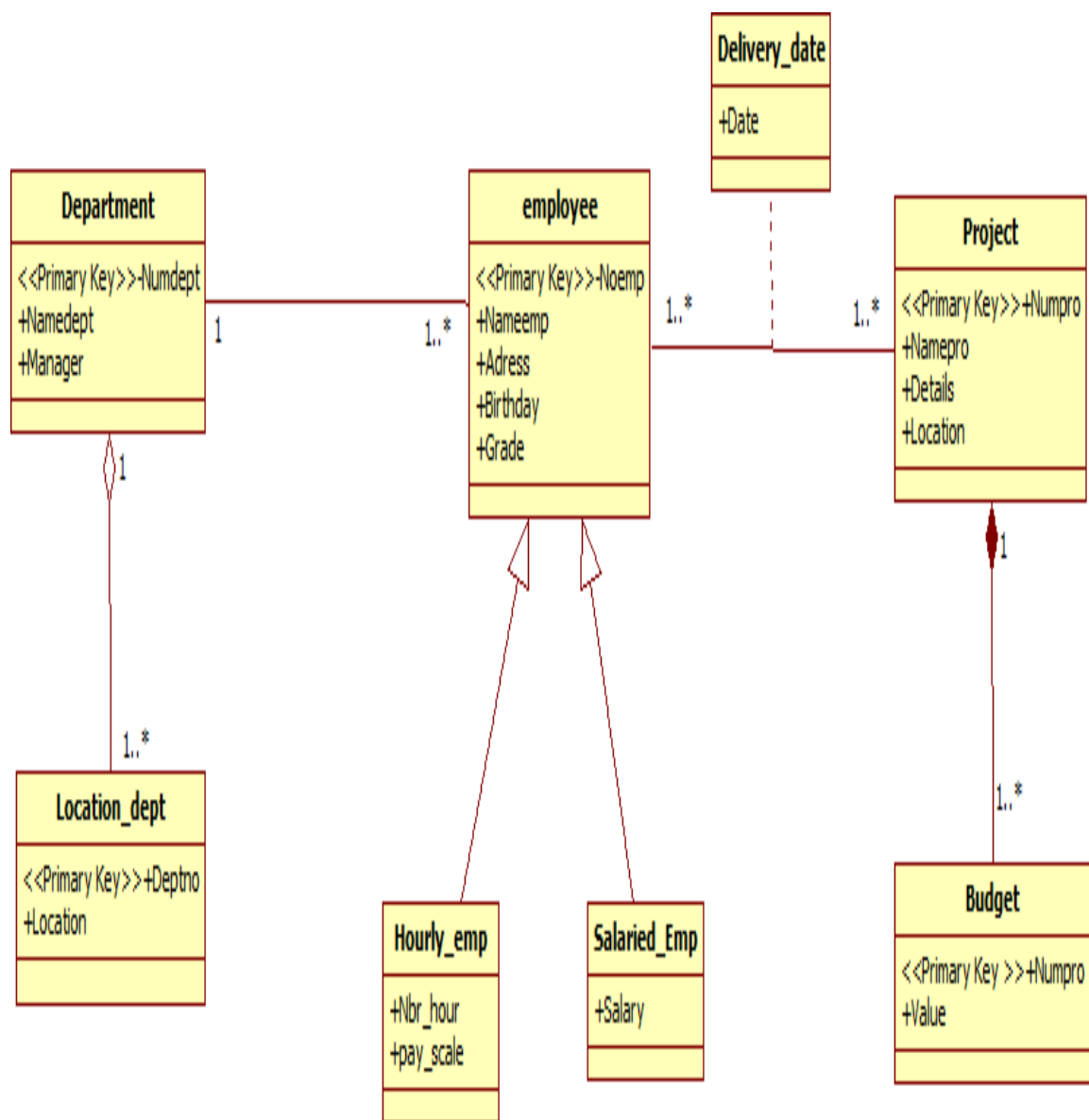


FIGURE 4.8: Class Diagram For Department

ACTIVITYDIAGRAM:

The activity diagram represents the workflow of the workforce analytics system by showing the sequence of activities involved from data input to the final output. It provides a clear understanding of how the system processes employee data step by step and how different operations are connected to each other. The diagram mainly focuses on the flow of control between various activities in the system.

The process starts when the user, such as an admin or HR manager, inputs employee-related data into the system. This data includes details like age, job role, salary, experience, performance level, and satisfaction. Once the data is provided, the system stores it in the My SQL database for further processing. After storing the data, the system performs data preprocessing, where it cleans the data by removing missing values, duplicates, and inconsistencies to ensure accuracy.

After preprocessing, the system moves to the data analysis stage, where Python is used to analyze the data and identify patterns and relationships. During this stage, the system examines factors such as employee performance, satisfaction, and attrition. Based on the analysis, the system then generates visual representations such as charts and graphs, which help in understanding the results more clearly.

Finally, the system generates reports that provide insights into workforce patterns, including performance analysis and attrition trends. These reports are then presented to the user for decision-making. The activity diagram clearly shows the flow from data input to output, including all intermediate steps such as storage, preprocessing, analysis, and visualization.

Overall, the activity diagram helps in understanding the complete workflow of the system and how each step is connected. It provides a simple and clear representation of the process, making it easier to understand the functioning of the workforce analytics system.

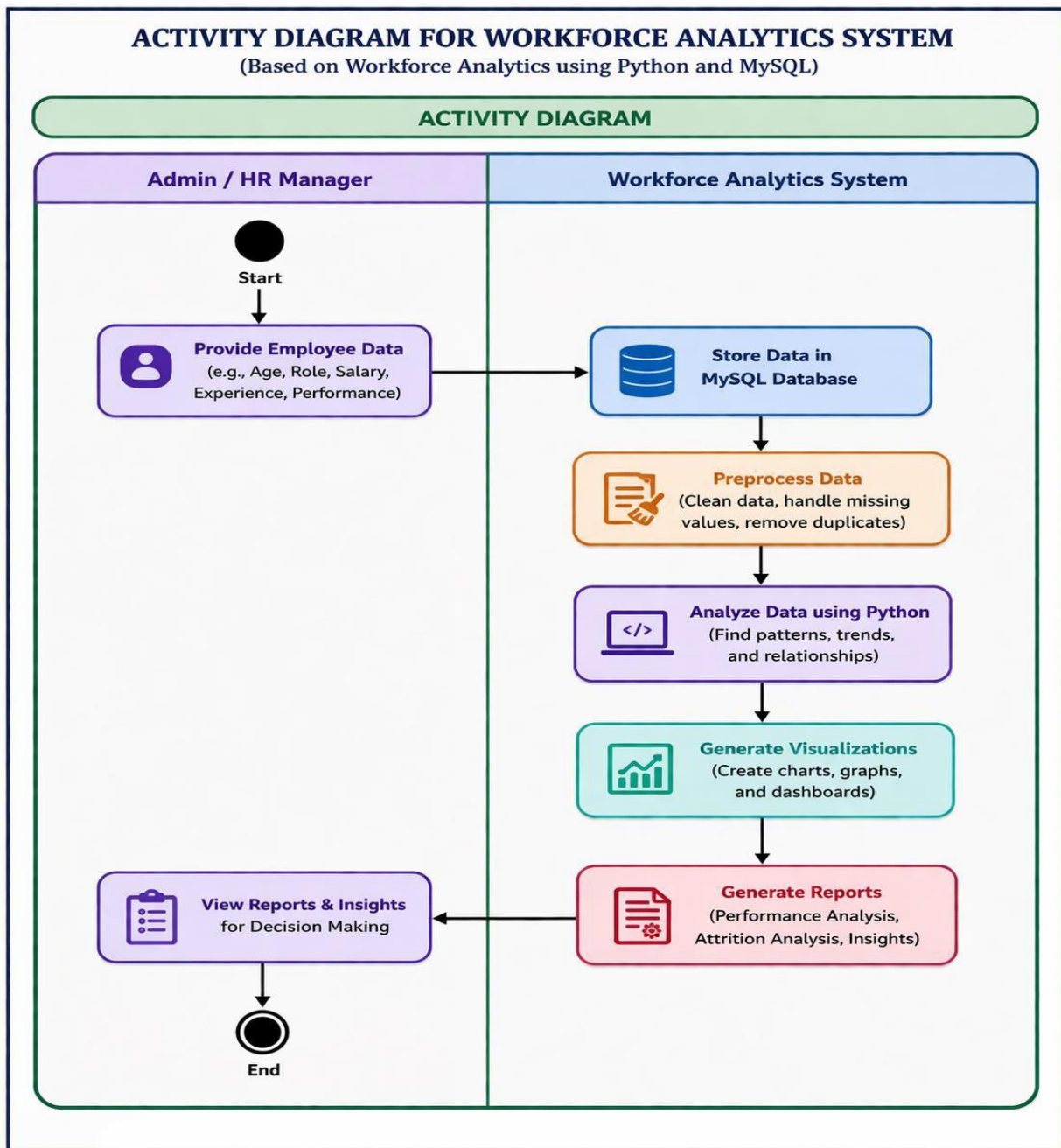


FIGURE 4.9: Activity Diagram For Workforce Analytics System

COLLABORATIONDIAGRAM:

The collaboration diagram represents how different components of the workforce analytics system interact with each other to complete the overall process. It focuses on the communication between various modules and shows how data is exchanged among them. This diagram helps in understanding the relationships between system components and how they work together to achieve the final output.

In this system, the interaction begins with the user, such as an admin or HR manager, who provides employee-related data to the system. The data is first received by the data collection module, which gathers and validates the input. The validated data is then passed to the database module, where it is stored in the MySQL database in a structured format.

Once the data is stored, the database module communicates with the data preprocessing module to retrieve the required data. The preprocessing module cleans the data by removing missing values, duplicates, and errors. After preprocessing, the cleaned data is forwarded to the data analysis module, where Python is used to analyze the data and identify important patterns and relationships.

The analyzed data is then sent to the visualization module, which creates graphical representations such as charts and graphs. These visual outputs help in better understanding the results. Finally, the visualization module communicates with the reporting module, which generates reports and presents them to the user.

The collaboration diagram clearly shows how each module interacts with others through communication links. It highlights the sequence of interactions and the flow of data between components. Overall, the diagram provides a clear view of how different parts of the workforce analytics system work together to process data and generate meaningful insights.

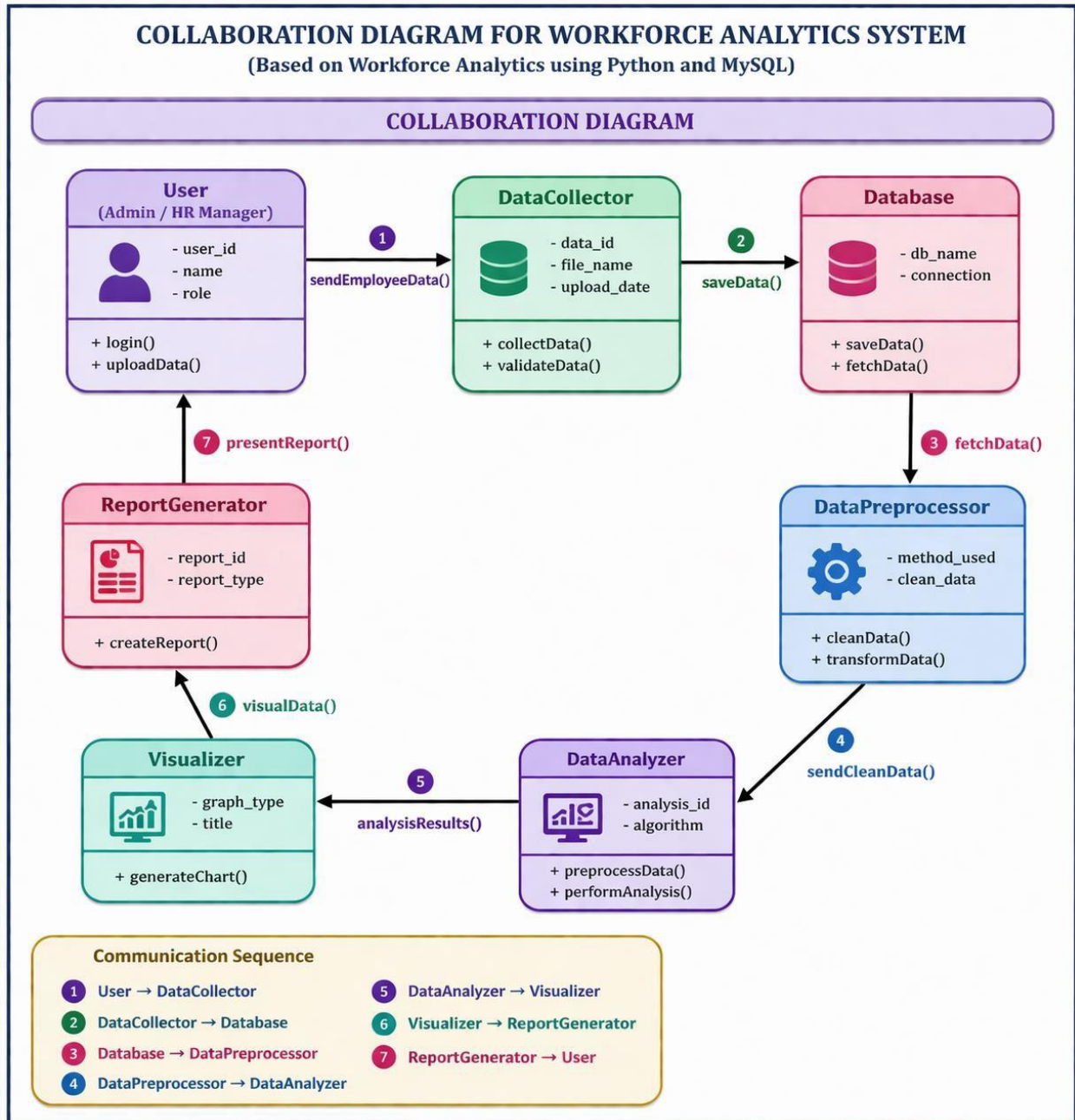


FIGURE 4.10: Collaboration Diagram For Workforce Analytics System

5. IMPLEMENTATION

IMPLEMENTATION:

The implementation of this project focuses on analyzing workforce data using Python and MySQL to understand employee patterns and organizational behavior. Initially, the dataset containing employee information such as age, job role, salary, experience, performance level, satisfaction, and attrition is collected and stored in a MySQL database. The database is designed in a structured format with tables that store employee details, ensuring easy access and proper data management.

After storing the data, Python is used to connect with the MySQL database using database connectors. The required data is retrieved from the database using SQL queries and loaded into Python for further processing. Data preprocessing is performed to handle missing values, remove duplicates, and convert the data into a suitable format for analysis. This step ensures that the dataset is clean and ready for accurate analysis.

Once the data is prepared, Python libraries such as Pandas and NumPy are used to perform data analysis. Various operations such as filtering, grouping, and aggregation are applied to identify patterns in employee data. The analysis focuses on understanding how different factors like job role, salary, and experience are related to employee performance and attrition. This helps in identifying trends and relationships within the organization.

Data visualization techniques are then applied using libraries like Matplotlib and Seaborn to represent the analyzed data in the form of graphs and charts. These visualizations help in clearly understanding employee distribution, performance levels, and attrition patterns. Graphical representations make it easier to interpret complex data and support better decision-making.

The system also allows comparison between different employee groups to identify which departments or roles have higher retention or turnover rates. This helps in understanding the impact of workplace conditions on employee stability. The insights generated from the analysis can be used by organizations to improve workforce planning and management strategies.

SAMPLECODE

```
import pandas as pd
import numpy as np

# CHECK DATATYPES
print(df.dtypes)

# CONVERT COLUMN NAMES TO LOWERCASE + UNDERSCORE FORMAT
df.columns = df.columns.str.lower().str.strip().str.replace(" ", "_")
df.columns

# CHECK NULL VALUES
print(df.isnull().sum())

# REMOVE DUPLICATES
df.drop_duplicates(inplace=True)

# FEATURE ENGINEERING - NEW COLUMNS
# Experience category
df['experience_category'] = pd.cut(
    df['total_working_years'],
    bins=[-1, 2, 5, 10, 20, 40],
    labels=['Fresher', 'Junior', 'Mid-Level', 'Senior', 'Expert']
)

# Income yearly
df['annual_income'] = df['monthly_income'] * 12
```

```
#Salaryefficiency
```

```
df['income_per_year_experience']=df['monthlyincome']/(df['totalworkingyears']+1)
```

```
# Distance category
```

```
df['distance_category']=pd.cut(  
    df['distancefromhome'],  
    bins=[0, 5, 15, 30],  
    labels=['Near','Moderate','Far']  
)
```

```
# Satisfaction score combined
```

```
df['overall_satisfaction_score']=(  
    df['environmentsatisfaction']+  
    df['jobsatisfaction'] +  
    df['relationshipsatisfaction']  
)
```

```
# Promotion readiness indicator
```

```
df['promotion_ready']=np.where(  
    (df['performancerating']>=3)&  
    (df['joblevel'] >= 2),  
    'Yes',  
    'No'
```



```

)

# Attrition risk score

df['attrition_risk_score']=(

    df['overtime'].map({'Yes':2,'No':0})+

    (4 - df['jobsatisfaction']) +

    (4- df['environmentsatisfaction'])

)

# Work stability score

df['work_stability_score']=(

    df['totalworkingyears'] -

    df['numcompaniesworked']

)

df.columns

#AnalysisStarts

#Promotion Ready Employees by Department

df.groupby('department')['promotion_ready'].value_counts()

#Work Stability by Age Group

df.groupby('agegroup')['work_stability_score'].mean()#High

Attrition Risk Employees

df[df['attrition_risk_score']>=5]

# SAVE CLEANED FILE

```

```

df.to_csv(r"D:\project\HR\proposed1.csv",index=False)

!pipinstallmysql-connector-pythonpymysqsqlalchemy

importpandasaspd

fromsqlalchemyimportcreate_engine from

urllib.parse import quote_plus

#Loadcleanedfile

df=pd.read_csv(r"D:\project\HR\proposed1.csv")

# MySQL connection

username = "root"

password="arfath123" host

= "127.0.0.1"

port="3306"

database="hrr_project"

engine = create_engine(

    f'mysql+pymysql://{username}:{password}@{host}:{port}/{database}'

)

#Upload toMySQL

```

```
df.to_sql("hr_data",engine,if_exists="replace", index=False)
```

```
#Verifyupload
```

```
pd.read_sql("SELECT*FROMhr_dataLIMIT5;", engine)
```

6. SYSTEM TESTING

System testing is an important phase in the development of the workforce analytics system, where the complete system is tested to ensure that it functions correctly and meets the project requirements. In this project, system testing is performed after integrating all modules such as data collection, data storage, preprocessing, data analysis, visualization, and reporting. The main objective of system testing is to verify that the system works as expected and produces accurate results when processing employee data.

The testing process begins by providing sample employee data to the system, including attributes such as age, job role, salary, experience, performance level, satisfaction, and attrition. The system is then tested to ensure that the data is correctly stored in the MySQL database and can be retrieved without any errors. After this, the preprocessing module is tested to check whether it properly handles missing values, removes duplicates, and cleans the dataset.

Further testing is carried out on the data analysis module to verify that Python correctly analyzes the data and identifies patterns and relationships among different factors. The visualization module is tested to ensure that graphs and charts are generated accurately and clearly represent the analyzed data. The reporting module is also tested to confirm that it produces meaningful reports that help in understanding workforce patterns.

In addition, the system is tested for performance and reliability by using different datasets to check how it handles large amounts of data. The system is also checked for accuracy to ensure that the results generated are correct and consistent. Error handling is tested to verify that the system can manage invalid inputs without crashing.

Overall, system testing ensures that all components of the workforce analytics system are working together properly and that the system provides reliable and accurate insights. This helps in delivering a high-quality system that supports better decision-making and workforce management.

Testing Principles:

Testing is an essential part of developing the workforce analytics system, as it ensures that the system functions correctly and produces accurate results. One of the main principles followed in this project is that testing should begin at the early stages of development. By testing each module such as data collection, storage, preprocessing, and analysis individually, errors can be identified and corrected at an early stage, reducing the overall complexity of the system.

Another important principle is that testing should be based on the project requirements. The system is tested to ensure that it meets all functional requirements, such as correctly storing employee data, analyzing it using Python, and generating accurate reports. This ensures that the system performs all intended operations without failure.

The principle of complete testing is also followed, where all possible scenarios are considered. The system is tested with different types of data, including valid, invalid, and missing values, to ensure that it can handle various situations effectively. This helps in improving the reliability and robustness of the system.

Error detection is another key principle, where the focus is on identifying defects in the system rather than proving that the system is error-free. Testing helps in finding bugs and issues in data processing, analysis, and visualization, which can then be corrected to improve system performance.

The principle of integration testing is applied to ensure that all modules of the system work together properly. After testing individual modules, they are combined and tested as a whole system to verify smooth data flow between MySQL and Python components.

Finally, testing should be continuous and systematic. The system is tested at different stages of development and updated whenever changes are made. This ensures that the system remains stable and performs efficiently throughout its lifecycle.

Overall, these testing principles help in developing a reliable, accurate, and efficient workforce analytics system that supports effective decision-making and organizational planning.

Test Cases:

Test Case1: Data Input Testing

- **Input:** Employee data (age, job role,salary, etc.)
- **Expected Output:** Data should be accepted without errors
- **Result:** Successfully stored

Test Case2: Data Storage Testing

- **Input:** Employee dataset
- **Expected Output:** Data should be stored correctly in MySQL
- **Result :** Data stored successfully

Test Case3: Data Retrieval Testing

- **Input:** Query to fetch employee data
- **Expected Output:** Correct data should be retrieved
- **Result:** Data retrieved accurately

Test Case4: Data Preprocessing Testing

- **Input:** Dataset with missing and duplicate values
- **Expected Output:** Clean data set after preprocessing
- **Result:** Missing values handled and duplicates removed

Test Case5: Data Analysis Testing

- **Input:** Clean employee dataset
- **Expected Output:** Correct analysis of patterns and trends
- **Result:** Analysis performed successfully

Test Case6: Visualization Testing

- **Input:** Analyzed data
- **Expected Output:** Graphs and charts generated
- **Result:** Visualizations displayed correctly

Test Case7: Report Generation Testing

- Successfully **Input:** Final analyzed data
- **Expected Output:** Reports generated with insights
- **Result:** Reports created

Test Case8: Performance Testing

- **Input:** Large dataset
- **Expected Output:** System should handle data efficiently
- **Result:** System performs without delay

Test Case9: Error Handling Testing

- **Input:** Invalid or incorrect data
- **Expected Output:** System should show error without crashing
- **Result:** Errors handled properly

Test Case10: Integration Testing

- **Input:** Full system workflow
- **Expected Output:** All modules work together smoothly
- **Result:** System works correctly

System Testing Positives Test Case:

Test CaseID	Test Case Description	\Input/Action	Expected Result
TC01	Add new employee details	Enter valid employee data(Age, Job Role, Salary, Experience, Performance, Satisfaction)	Employee record is saved successfully in the system
TC02	View employee data	Select employee from the employee list	System displays correct employee information
TC03	Search employee by job role	Enter valid job role(e.g., Manager, Developer)	System shows employees with that job role
TC04	Generate workforce distribution report	Click "Generate Report" option	System displays report showing Employee distribution across departments
TC05	Analyze employee performance	Select performance analysis option	System shows performance levels of employees correctly
TC06	View attrition analysis	Select attrition analysis option	System displays turnover/retention statistics
TC07	Update employee information	Edit valid employee details and save	System updates the employee record successfully
TC08	View employee satisfaction data	Select satisfaction report option	System shows satisfaction levels Of employees
TC09	Generate experience-based report	Select report based on work experience	System displays employee experience distribution
TC10	Log into system	Enter valid user name and password	User successfully logs into the system

System Testing Negative Test Case:

Test Case ID	Test Case Description	Invalid Input/Action	Expected Result
TC01	Add employee with empty fields	Leave required fields(Age, Job Role, Salary) blank	System should display error message and not save data
TC02	Enter invalid age	Enter negative age or text (e.g., -25 or ABC)	System should reject input and show validation message
TC03	Enter incorrect salary format	Enter letters instead of numbers (e.g., "Ten Thousand")	System should display error and ask for numeric value
TC04	Search employee with invalid job role	Enter job role that does not exist	System should display "No records found" message
TC05	Login with wrong password	Enter in correct user name or password	System should deny access and show login error
TC06	Update employee with invalid data	Enter in correct experience value (e.g., -5 years)	System should not update record and show error message
TC07	Generate report without data	Try to generate report when no employee data exists	System should show message like "Nodata available"
TC08	Enter extremely large numbers	Enter very large salary value (e.g., 9999999999)	System should validate and restrict input
TC09	Submit form without selecting job role	Leave job role drop down unselected	System should prompt user to select a role
TC10	Enter special characters in text fields	Enter symbols like @#\$% in name field	System should reject invalid characters

7. SCREENSHOTS

```
: # Analysis Starts
#Promotion Ready Employees by Department
df.groupby('department')['promotion_ready'].value_counts()
```

```
: department      promotion_ready
Human Resources   No              33
                  Yes             30
Research & Development Yes       527
                  No            434
Sales             Yes            370
                  No             76
Name: count, dtype: int64
```

```
: #Work Stability by Age Group
df.groupby('agegroup')['work_stability_score'].mean()
```

```
: agegroup
18-25    1.813008
26-35    5.521452
36-45    9.655983
46-55   16.654867
55+     16.382979
Name: work_stability_score, dtype: float64
```

```
6
7  -- 1 Total employees
8 • SELECT COUNT(*) AS total_employees FROM hr_data;
9
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	total_employees
▶	1470

FIGURE 7.1: Work Stability By Age Group

```

10      -- 2 Attrition count
11 •    SELECT attrition, COUNT(*)
12      FROM hr_data
13      GROUP BY attrition;

```

Result Grid			Filter Rows:	Expo
	attrition	COUNT(*)		
▶	Yes	237		
	No	1233		

```

15      -- 3 Department wise employees
16 •    SELECT department, COUNT(*)
17      FROM hr_data
18      GROUP BY department;

```

Result Grid			Filter Rows:	Expo
	department	COUNT(*)		
▶	Research & Development	961		
	Sales	446		
	Human Resources	63		

FIGURE 7.2: Show Employees Attrition

```

20      -- 4 Average salary by department
21 •    SELECT department, AVG(monthlyincome)
22      FROM hr_data
23      GROUP BY department;

```

Result Grid | Filter Rows: | Export:

	department	AVG(monthlyincome)
▶	Research & Development	6281.2529
	Sales	6959.1726
	Human Resources	6654.5079

```

25      -- 5 Highest paying job roles
26 •    SELECT jobrole, AVG(monthlyincome) AS avg_salary
27      FROM hr_data
28      GROUP BY jobrole
29      ORDER BY avg_salary DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: ☐

	jobrole	avg_salary
▶	Manager	17181.6765
	Research Director	16033.5500
	Healthcare Representative	7528.7634
	Manufacturing Director	7295.1379
	Sales Executive	6924.2791
	Human Resources	4235.7500
	Research Scientist	3239.9726
	Laboratory Technician	3237.1699
	Sales Representative	2626.0000

FIGURE 7.3: Average Salary By Department

```

31 -- 6 Attrition by department
32 • SELECT department, attrition, COUNT(*)
33 FROM hr_data
34 GROUP BY department, attrition;
35

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	department	attrition	COUNT(*)
▶	Research & Development	Yes	133
	Sales	No	354
	Sales	Yes	92
	Research & Development	No	828
	Human Resources	Yes	12
	Human Resources	No	51

```

36 -- 7 Overtime impact on attrition
37 • SELECT overtime, attrition, COUNT(*)
38 FROM hr_data
39 GROUP BY overtime, attrition;

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	overtime	attrition	COUNT(*)
▶	No	Yes	110
	No	No	944
	Yes	Yes	127
	Yes	No	289

FIGURE 7.4: Show Attrition By Department

```

41      -- 8 Gender distribution
42      • SELECT gender, COUNT(*)
43      FROM hr_data
44      GROUP BY gender;

```

Result Grid			Filter Rows:
	gender	COUNT(*)	
▶	Male	882	
	Female	588	

```

46      -- 9 Average experience by department
47      • SELECT department, AVG(totalworkingyears)
48      FROM hr_data
49      GROUP BY department;

```

Result Grid			Filter Rows:	Export:	Wrap Cell C
	department	AVG(totalworkingyears)			
▶	Research & Development	11.3424			
	Sales	11.1054			
	Human Resources	11.5556			

```

51      -- 10 Salary slab distribution
52      • SELECT salaryslab, COUNT(*)
53      FROM hr_data
54      GROUP BY salaryslab;

```

Result Grid			Filter Rows:
	salaryslab	COUNT(*)	
▶	Upto 5k	749	
	5k-10k	440	
	10k-15k	148	
	15k+	133	

FIGURE 7.5: Gender Distribution

```

56  -- 11 Job satisfaction by job role
57  •  SELECT jobrole, AVG(jobsatisfaction)
58     FROM hr_data
59     GROUP BY jobrole;

```

Result Grid			Filter Rows:	Export:	Wrap Cel
	jobrole	AVG(jobsatisfaction)			
▶	Laboratory Technician	2.6911			
	Sales Representative	2.7349			
	Research Scientist	2.7740			
	Human Resources	2.5577			
	Manufacturing Director	2.6828			
	Sales Executive	2.7546			
	Healthcare Representative	2.7863			
	Research Director	2.7000			
	Manager	2.7059			

FIGURE 7.6: Job Satisfaction By Job Role


```

61  -- 12 Employees with highest salary hike
62  • SELECT empid, monthlyincome, percentsalaryhike
63  FROM hr_data
64  ORDER BY percentsalaryhike DESC;

```

Result Grid			
		Filter Rows:	
		Export:	Wrap Cell Content:
	empid	monthlyincome	percentsalaryhike
▶	RM854	2552	25
	RM893	1859	25
	RM1308	3591	25
	RM121	2613	25
	RM435	10648	25
	RM1076	11244	25
	RM1199	5368	25
	RM1088	2308	25
	RM677	4014	25
	RM1433	13744	25
	RM252	10938	25
	RM1373	5151	25

hr_data 12 ×

Output

FIGURE 7.7: Monthly Income


```

66      -- 13 Most experienced employees
67 •    SELECT empid, totalworkingyears
68      FROM hr_data
69      ORDER BY totalworkingyears DESC;

```

Result Grid   Filter Rows: Export 

	empid	totalworkingyears
▶	RM127	40
	RM596	40
	RM099	38
	RM271	37
	RM446	37
	RM086	37
	RM1302	37
	RM895	36
	RM1185	36
	RM188	36
	RM1117	36
	RM402	36

```

71      -- 14 Attrition by marital status
72 •    SELECT maritalstatus, attrition, COUNT(*)
73      FROM hr_data
74      GROUP BY maritalstatus, attrition;

```

Result Grid   Filter Rows: Export: 

	maritalstatus	attrition	COUNT(*)
▶	Single	Yes	120
	Single	No	350
	Divorced	No	294
	Married	No	589
	Married	Yes	84
	Divorced	Yes	33

FIGURE 7.8: Total Working Hours Per Year

```

76  -- 15 Education field wise salary
77  • SELECT educationfield, AVG(monthlyincome)
78  FROM hr_data
79  GROUP BY educationfield;

```

Result Grid   Filter Rows: Export:  Wrap C

	educationfield	AVG(monthlyincome)
▶	Life Sciences	6463.2888
	Medical	6510.0366
	Marketing	7348.5849
	Technical Degree	5758.3030
	Other	6071.5488
	Human Resources	7241.1481

FIGURE 7.9: Employees Monthly Income

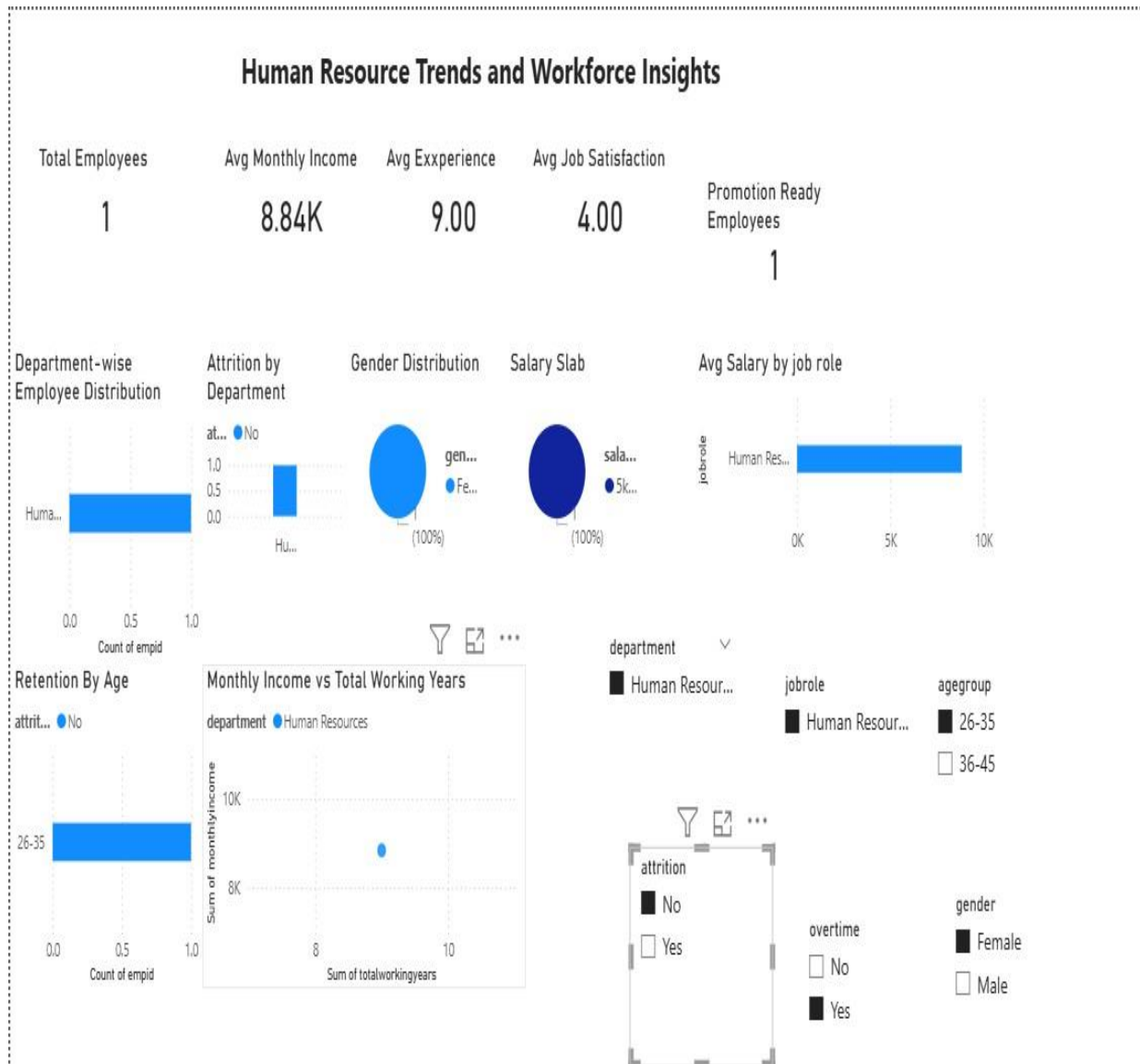


FIGURE 7.10: Dashboard Visualization On Power BI

8. CONCLUSION & FUTURE SCOPE

CONCLUSION:

The project on Human Resource Trends and Workforce Insights provides a comprehensive understanding of how employee data can be transformed into meaningful information through a structured analytics approach. By analyzing various aspects such as employee distribution, attrition patterns, salary structure, experience levels, and satisfaction metrics, the project presents a clear view of workforce behavior within an organization. From the outputs and analysis, it is evident that workforce distribution is not uniform and varies across departments and job roles. Certain departments have higher employee concentration, indicating their operational importance. Attrition analysis highlights that employee turnover exists in specific areas and is influenced by multiple factors such as job role, workload, and satisfaction levels. Salary distribution shows a structured variation based on roles and responsibilities, while experience analysis reflects a mix of employees with different tenure levels, contributing to both stability and growth within the organization.

The analysis of satisfaction and performance reveals that employee engagement plays a significant role in workforce stability. Departments or groups with better satisfaction levels tend to show stronger retention patterns. Similarly, factors like overtime provide additional context in understanding employee behavior, showing how workload can impact attrition. These observations collectively help in identifying patterns that are essential for workforce planning. The project demonstrates several benefits in terms of data handling and analysis. The use of pandas and numpy ensures that raw data is cleaned and structured effectively. Storing data in SQL allows efficient querying and faster retrieval of information. The use of matplotlib and Power BI improves the clarity of analysis by presenting data in visual formats. The integration of these tools creates a smooth workflow from raw data to final insights, reducing manual effort and improving accuracy.

FUTURE SCOPE:

The future scope of workforce analytics is very promising as organizations are increasingly adopting advanced technologies to improve employee management and organizational performance. With the rapid growth of data generation, future systems will focus more on real-time data analysis, allowing organizations to monitor employee performance and workplace conditions instantly. This will help management take quick and effective decisions based on current data rather than relying only on past records. In the coming years, the integration of artificial intelligence and machine learning with workforce analytics is expected to increase significantly. These technologies will help in building more accurate predictive models to identify employee attrition, performance trends, and workforce requirements. Organizations will be able to predict employee behavior in advance and take preventive actions to improve retention and satisfaction. This will reduce costs related to employee turnover and improve overall efficiency.

Another important area of future development is the use of advanced data visualization tools. Interactive dashboards and real-time reporting systems will provide better insights into workforce data. Managers will be able to understand complex information easily through visual representations, which will improve communication and decision-making across the organization. Enhanced visualization techniques will make workforce analytics more user-friendly and accessible. The integration of workforce analytics systems with cloud technologies is also expected to grow. Cloud-based solutions will allow organizations to store and process large amounts of data securely and efficiently. This will make workforce analytics systems more scalable and flexible, especially for large organizations with distributed teams. Cloud platforms will also support remote access, enabling managers to analyze data from anywhere.

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